

Creating a Mesh from LIDAR and DSLR data (OR: How to Use Reality Capture and Cyclone Register 360)

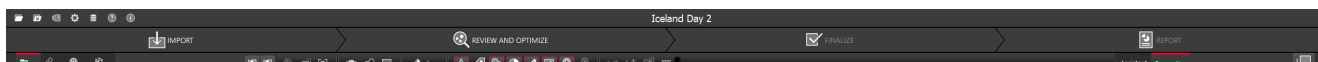
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Nicholas Anastas edited this page 12 days ago · 13 revisions

Generating e57 Files (using Cyclone REGISTER 360)

Option 1: Start from Scratch

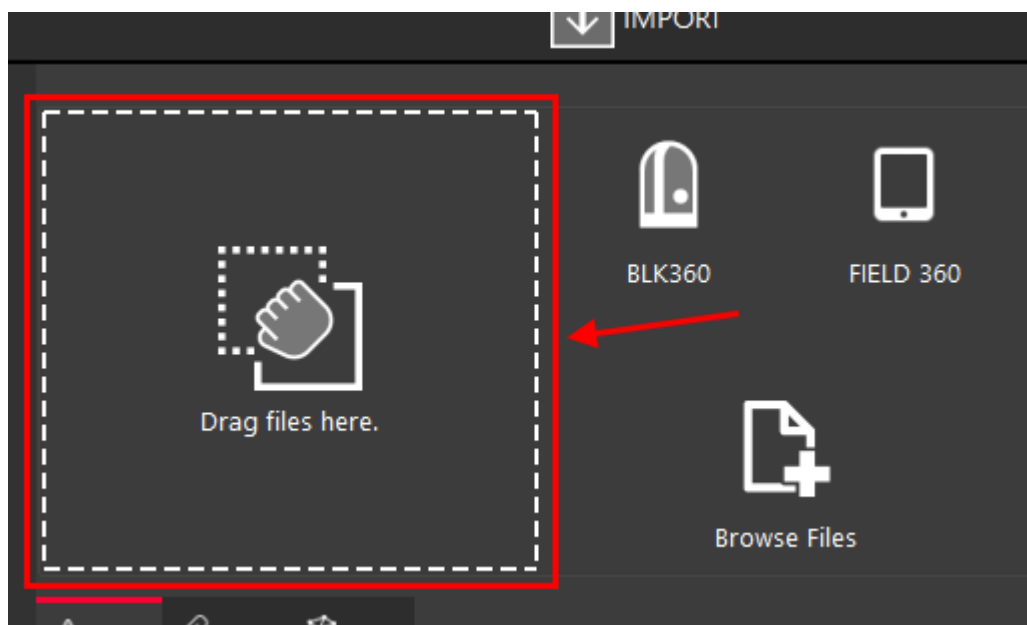
Cyclone REGISTER 360's (Cyclone) process of aligning a set of laser scan inputs and exporting them in a format usable by Reality Capture (RC) has 4 main steps: **Import**, **Review and Optimize**, **Finalize**, and **Report**. These steps, along with your progress through them, are shown at the top of the screen in Cyclone:



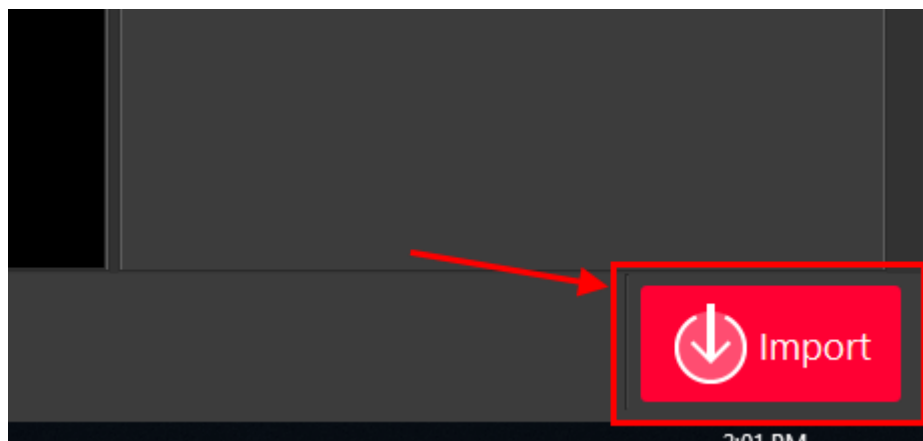
Import

In the import stage, simply drag and drop all of your laser scans (.blk files were used during the writing of this document) into the 'Drag files here' box:

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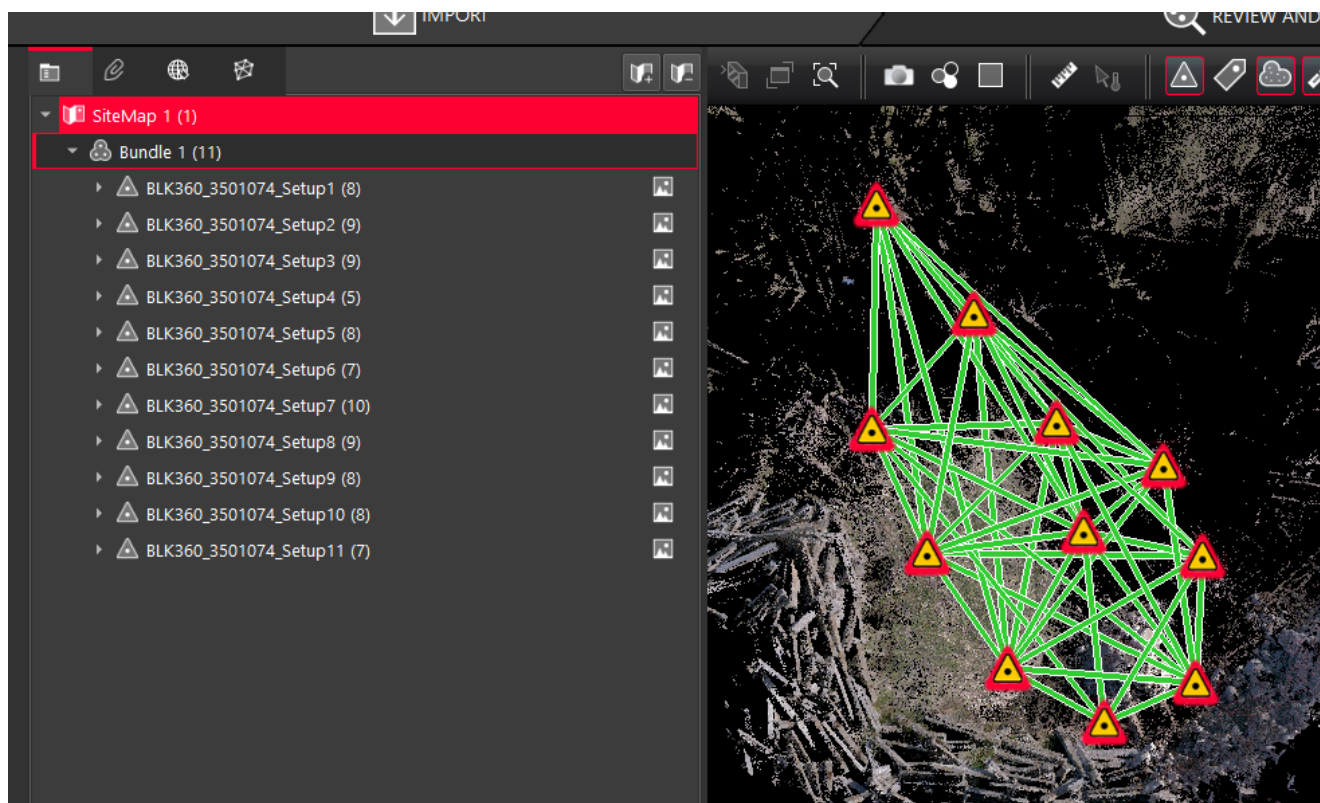
Once you've done that, simply click the 'Import' button in the lower right corner of the screen (and wait for the import process / loading bar to complete) and you'll be finished with this step:



Review and Optimize

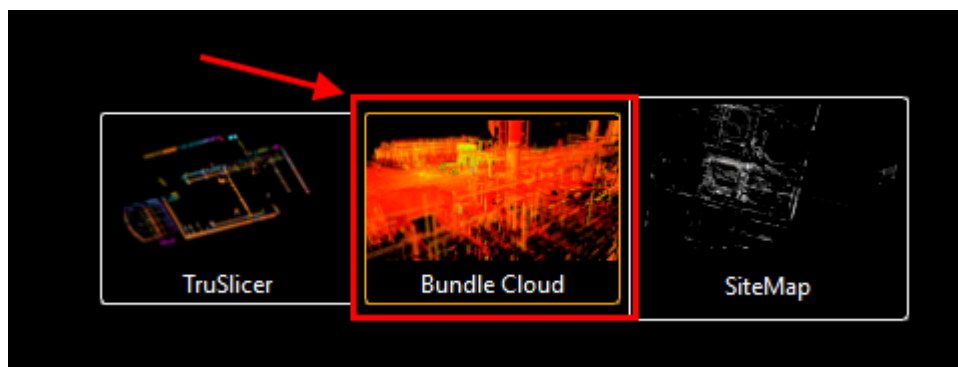
In an ideal world, Cyclone will have found links between all of your laser scan files (happens automatically during the import process) and there won't be a whole lot for you to do in this step. If that's the case, your 'Project' and 3D 'SiteMap' windows should look something like this:

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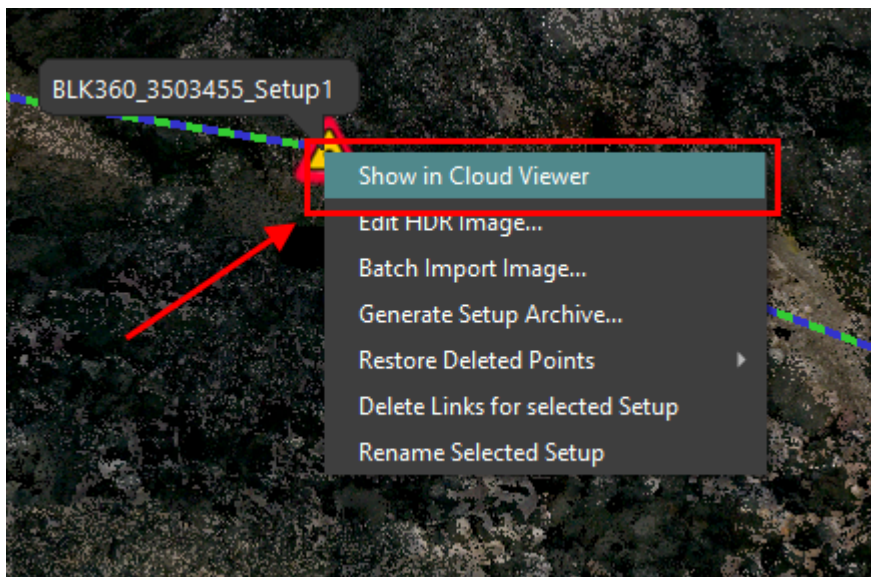
Cyclone will place laser scans that have been 'linked' into a common bundle. If two scans have been 'Linked', this means that Cyclone has discovered a relative alignment that satisfies a certain 'Target Error' limit (the error limit can be adjusted in the 'Links' tab of the 'Settings' dialogue).

Once you have all of your scans bundled in the way you want time, you can in this step choose to edit (e.g. select and delete cloud points) your scan data. To edit a scan, you'll likely want to be in 'cloud view' mode. To do this, either (with your bundle selected in the 'Project' window) hover over the 'SiteMap' box in the lower right hand corner of the 3D view and click the 'Bundle Cloud' button that appears:

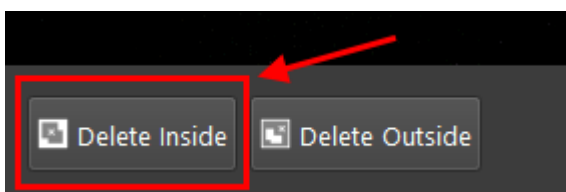


Or right click on any of the scan locations and choose 'Show in Cloud Viewer':

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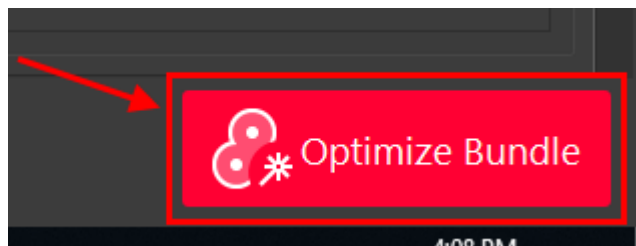
In this view mode, you can do handy things like select and delete points that you'd rather not have in your point clouds (e.g. scans of the people/tripods who actually created the scans). To delete a set of points, turn on the 'Point Selection' tool (hotkey 'F'), select your desired points, and click the 'Delete Inside' button that appears in the bottom left of the screen.



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A handy tip for selecting and deleting points: holding the 'ctrl' key will allow you to move the the camera around in whatever control scheme you have set (e.g. Orbit, Panorama).

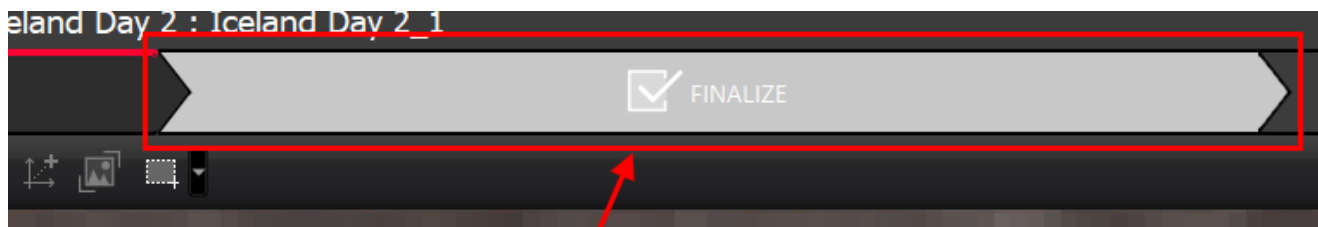
Once you're satisfied with the state of your bundles and point clouds, click (with your bundle selected in the 'Project' window) the 'Optimize Bundle' button in the lower left corner of the screen.



You should now be ready for the next step.

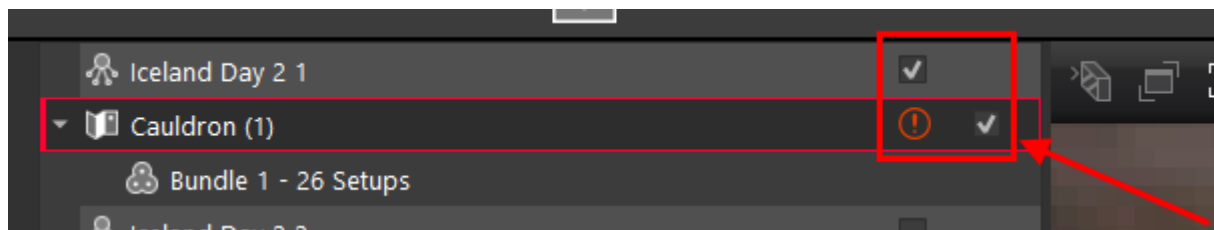
Finalize

To begin this step, you just need to click the 'FINALIZE' tab at the top of the screen:



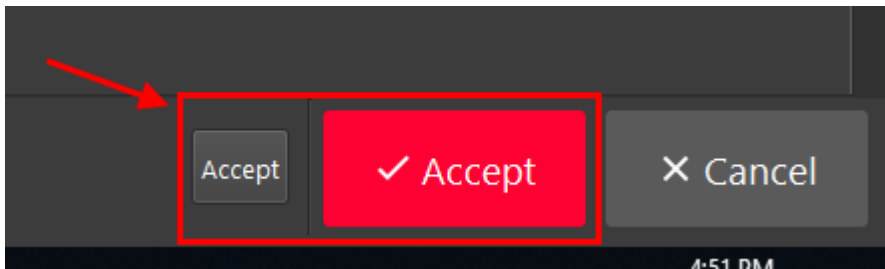
Note that for this tab to be click-able, you need to have at least one Sitemap whose child scans are all in one bundle (i.e. all of the scans are linked).

In this step, there's not a lot for you to do (probably). If you like the way things look, first make sure the Sitemap you'd like to export is checked in the 'Project' window:



and then just click the 'Accept' button in the lower right corner of the screen. Then click the *other* 'Accept' button that's just turned red on account of your first 'Accept' click (accepting twice makes the result twice as good!):

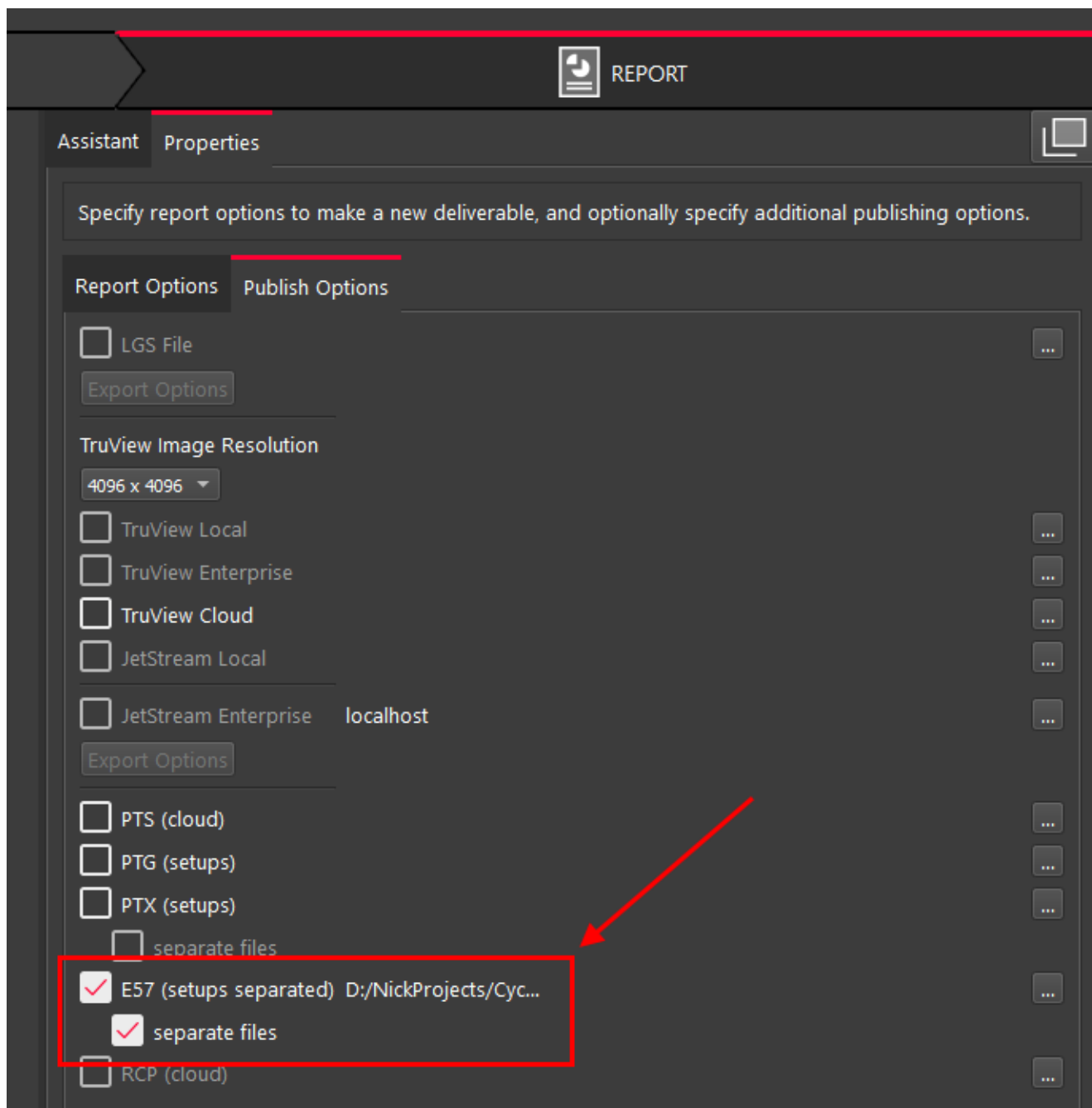
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Report

We've finally made it to the 'Report' step, which is where we'll actually be exporting the aligned .e57 files. To do this, navigate to the 'Publish Options' tab nested inside the 'Properties' tab. Near the bottom of the window, you'll find a checkbox with label 'E57 (as one file)'. Check that box (this will open a dialogue where you'll choose a file path and name for the export; don't worry that the button says 'save' - no saving will actually happen until we publish), and then also check the 'separate files' box beneath it.

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And now we're ready to export! Click the 'Publish' button in the lower right corner of the screen:



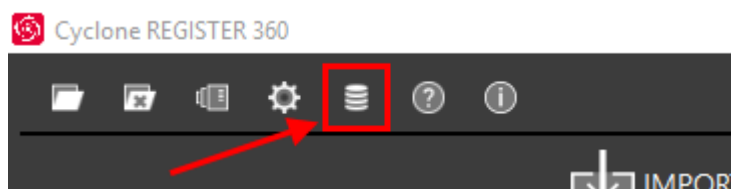
wait for the publishing to complete, and then have fun with your brand new set of aligned .e57 files!

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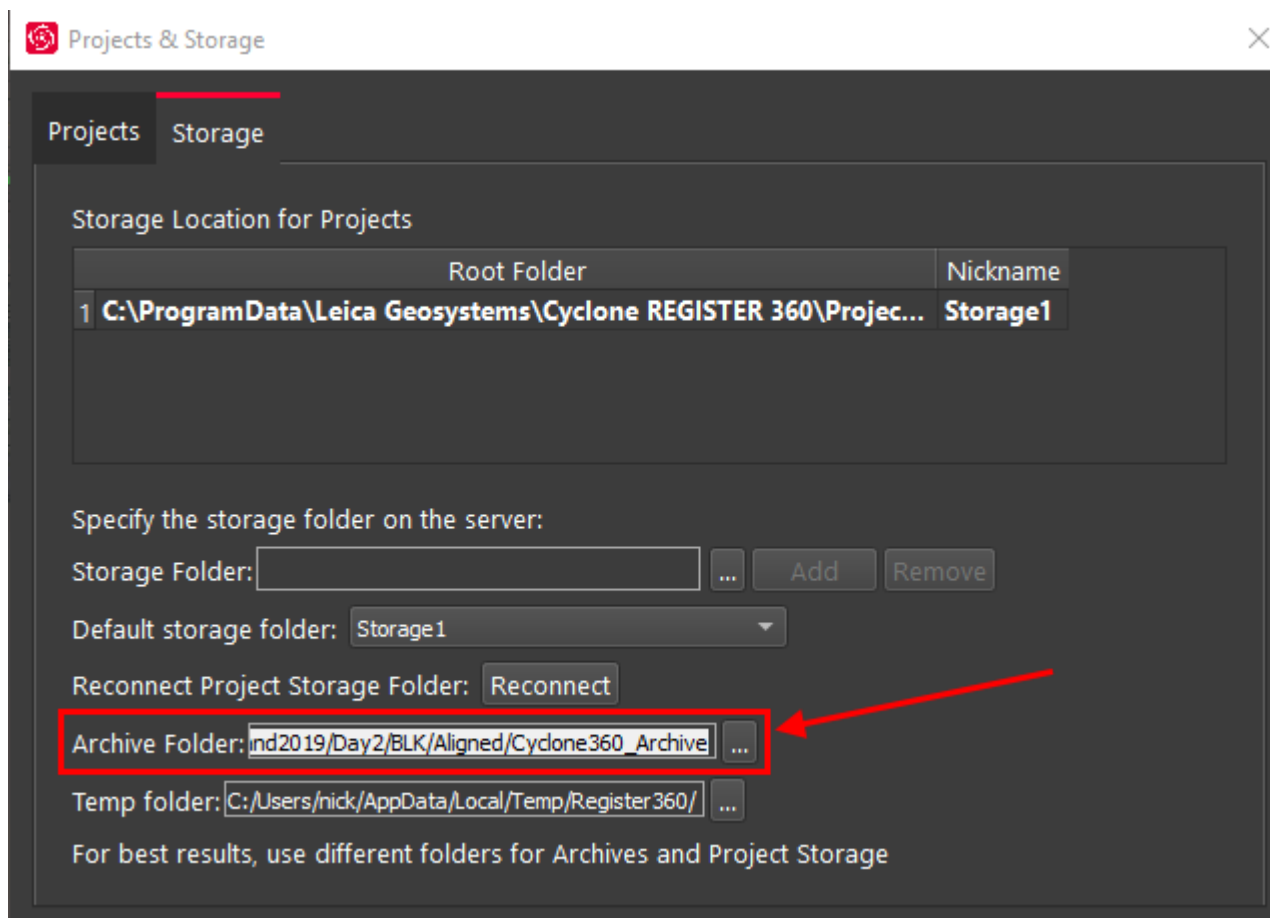
Option 2: Start from an Archive

If you're starting from a Reality Capture archive file (.raf file extension), you'll just want to import that archive and then follow whatever directions you need to from the 'Option 1: Start from Scratch' section above.

To import an archive file, open the 'Projects and Storage' dialogue by clicking the corresponding button (pancakes) on the upper left side of the screen:

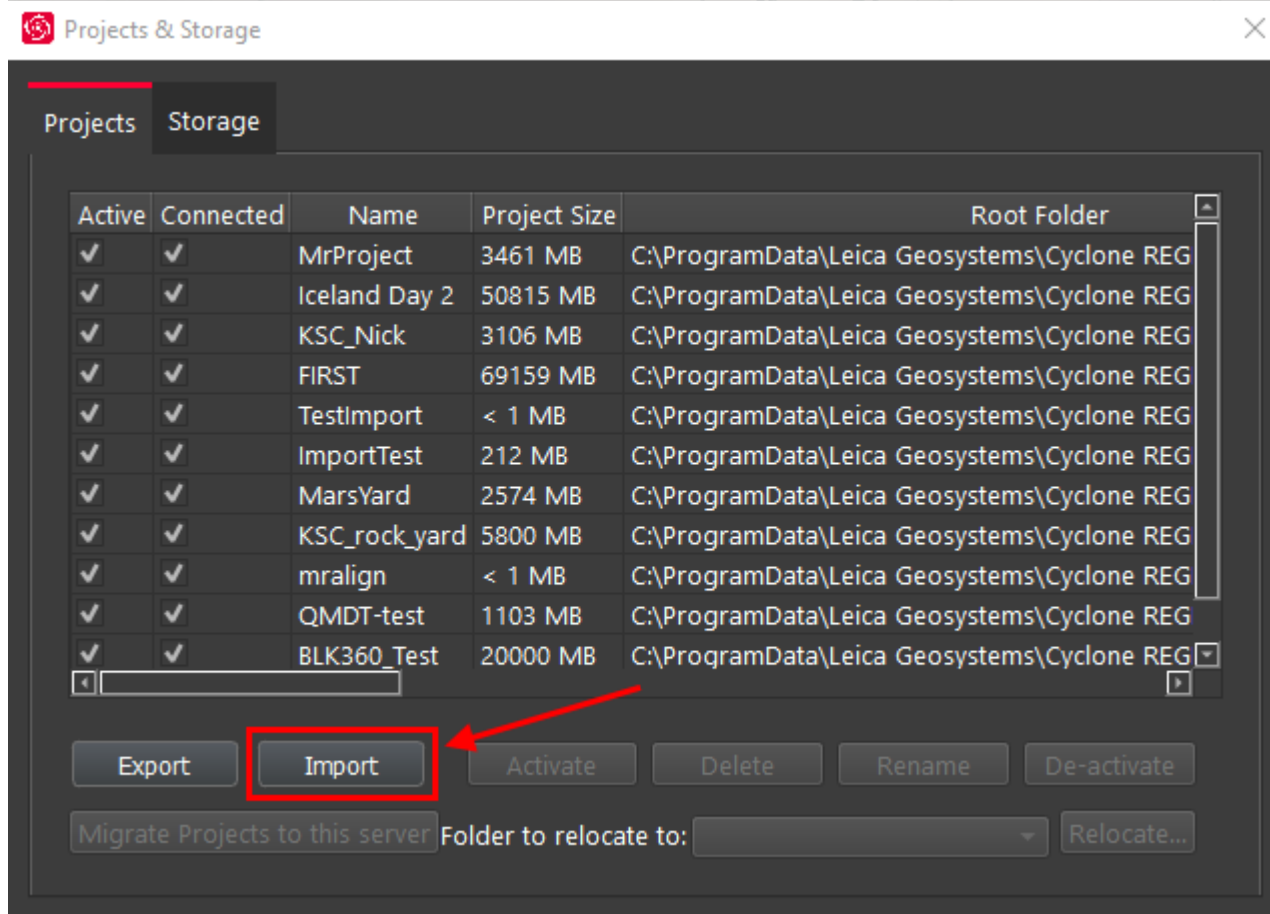


Next, you'll want to place the archive file in the Cyclone-designated 'Archive Folder' the location of this folder can be set on the 'Storage' tab of the 'Projects and Storage' dialogue:



With your .raf file now where you want it, you can complete the import process simply by clicking the 'Import' button on the 'Projects' tab of the 'Projects and Storage' dialogue:

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And you're all done! You can open your freshly-imported project in the exact same way as you would any traditionally created project (i.e. using the 'Open Project' dialogue).

Additional References

- [Cyclone 360 'Quick Start Guide'](#)

Creating a Textured Mesh (using Reality Capture)

Project Creation

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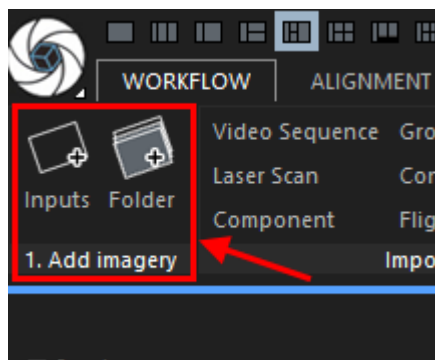
It's a good idea to create one RealityCapture (RC) project per set of inputs, or per set of images taken around the same time and having a substantial amount of overlap. RC generally does well with 'automatically' aligning inputs (i.e. pushing the 'align images' button and having things just work) that are grouped in this way. The basic idea here is to create one RC project per group of these 'similar' inputs, create and export a single RC component from each of these projects, and then merge all of the exported sub-components into one final super-component (which can then be meshed and textured).

Adding Inputs

For the purposes of this document, we'll say that RC accepts three main input types: images, laser scans, and RC components (other input types like videos aren't covered here). Images and laser scans come entirely from outside of the RC environment, while the RC components (as suggested by the name) were created in and exported from some other external RC project.

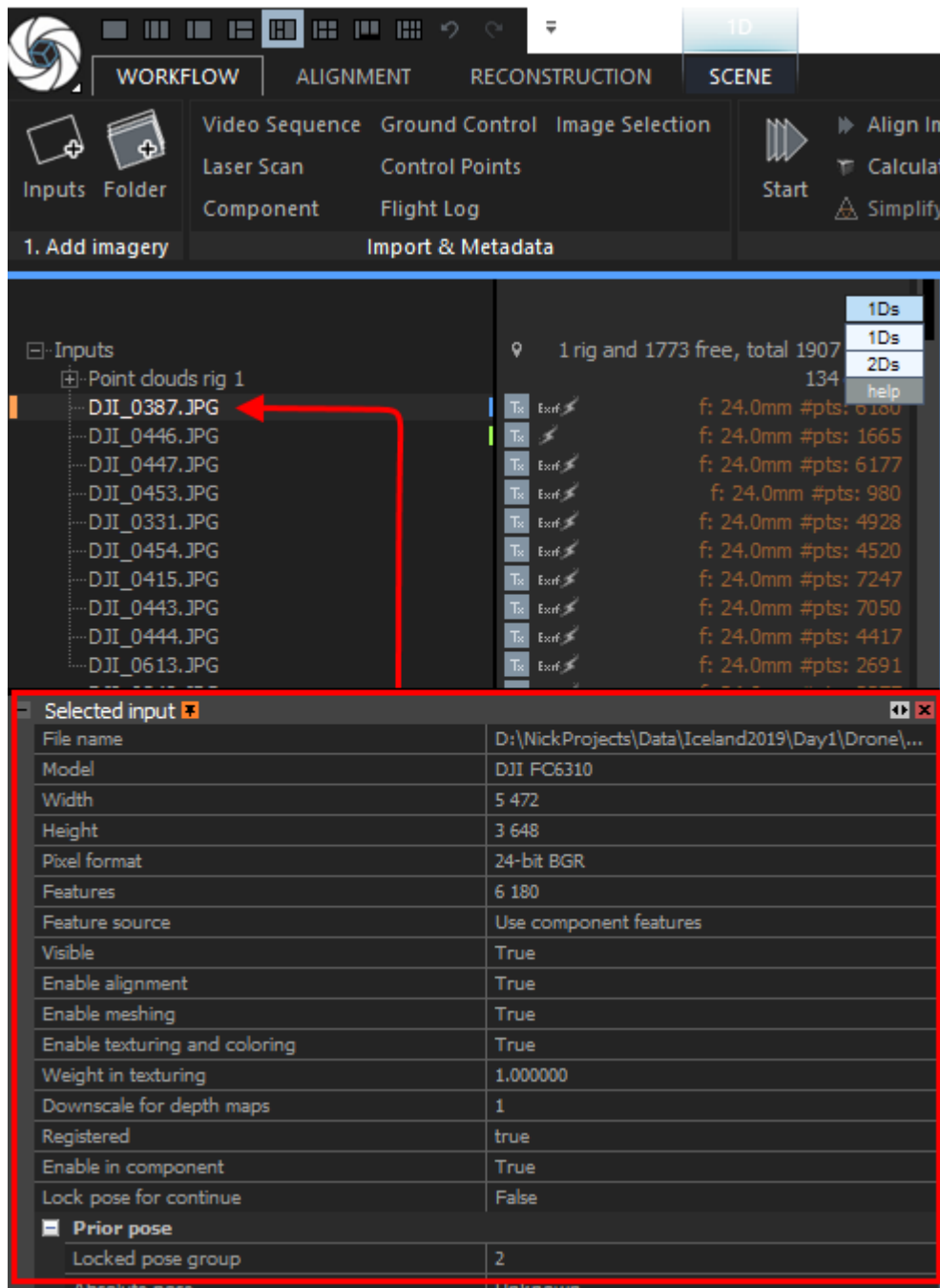
Images

Under the 'WORKFLOW' tab, click either the 'Inputs' button (to open an explorer window and select a specific set of images to import) or the 'Folder' button (to import all images in a specific folder).



No real options on the import - just click the button and you're done. RC will automatically import any relevant Exif metadata associated with an image, which can be viewed by clicking on the image in the '1Ds' window.

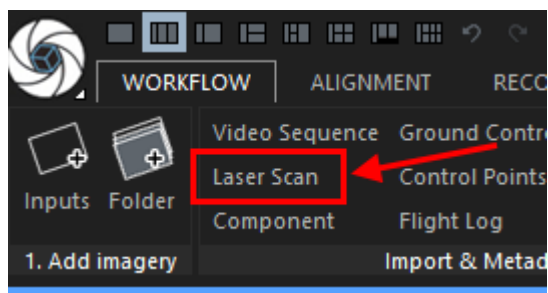
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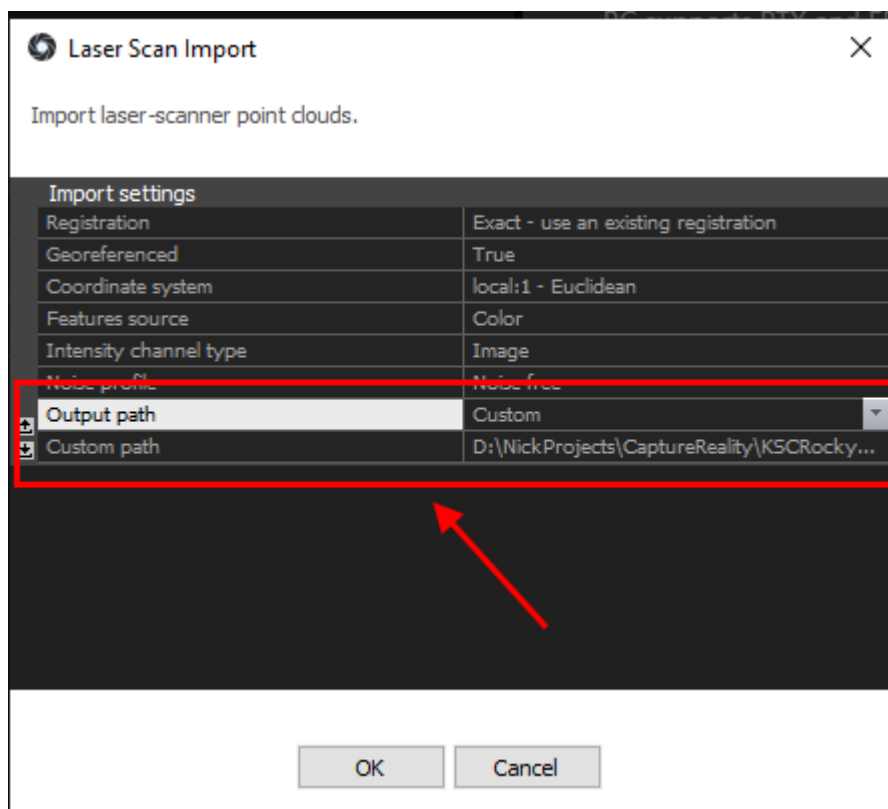
Laser Scans

At the time of this writing, RC supports the import of two laser scan types: PTX and E57. You can import files of these types by clicking the 'Laser Scan' button and selecting the files you want.

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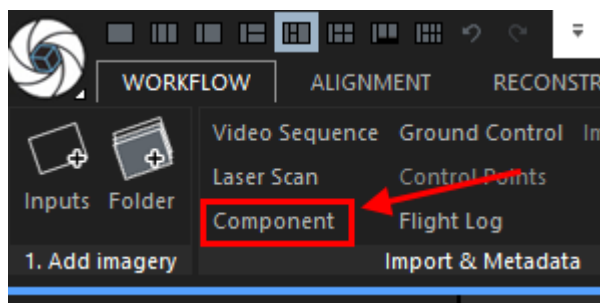
When RC imports laser scan files, it will convert them to an LSP file format. These LSP files are actually saved to disk - so be mindful of this when choosing an 'output path' in the 'Laser Scan Import Settings' dialogue.



Leaving the other import settings as their default values seems to give good results.

Components

To import a RC component (.rcalign file extension), simply click the 'Component' button on the 'WORKFLOW' tab and then navigate to and select the component you'd like to import.

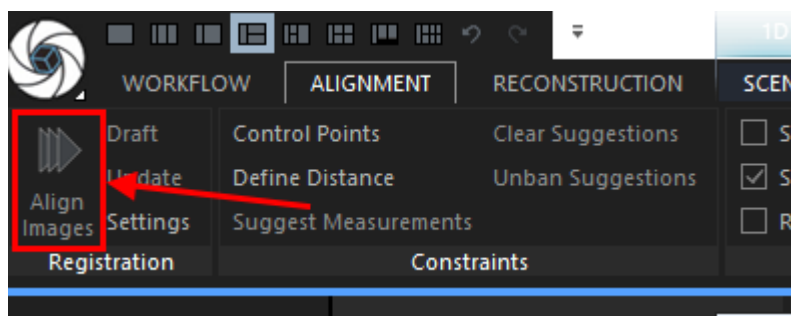


There are no import settings to choose from here, so just select and you're done.

When you import a component, all of the inputs associated with that component will also be imported; these imported inputs will have their 'Feature source' set to 'Use component features' ('Use all image features' is the default). With this 'Feature source' set, RC will use re-align the inputs with the points used in the alignment of the imported component.

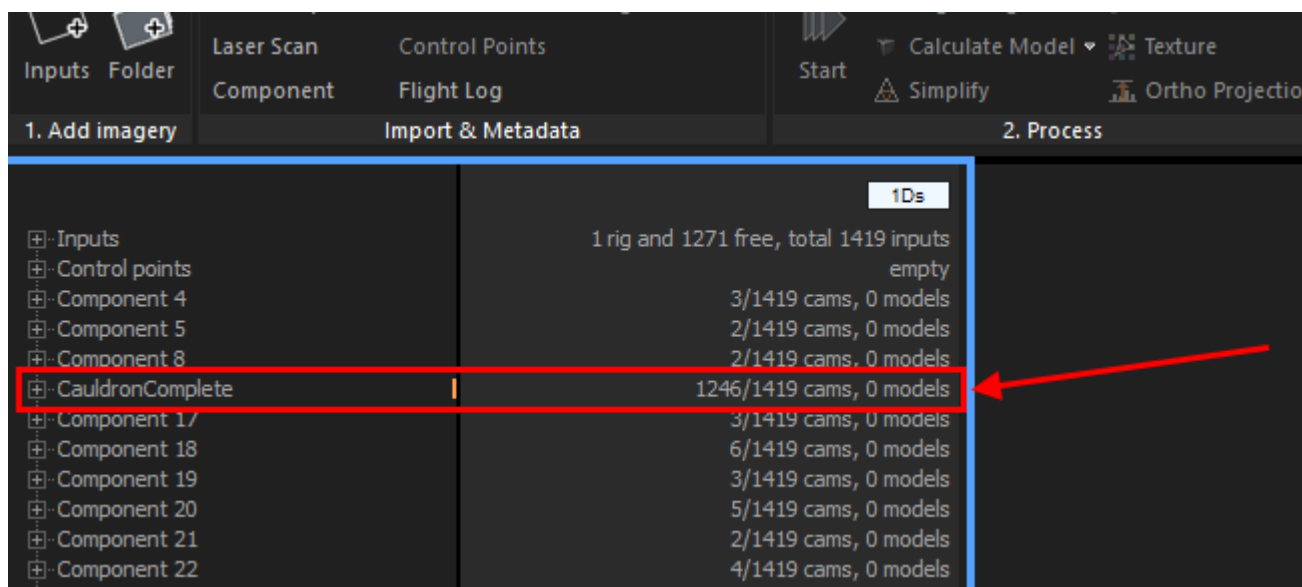
Alignment

Alignment is the most important and also (sometimes) the most difficult/finicky step. Here, RC is finding matching visual elements (tie points) between pictures, and using those to determine the poses from which all of the input images were taken. If your set of images is 'good', all you *should* need to do is click on the large 'Align Images' button on the leftmost side of the 'ALIGNMENT' tab/ribbon:



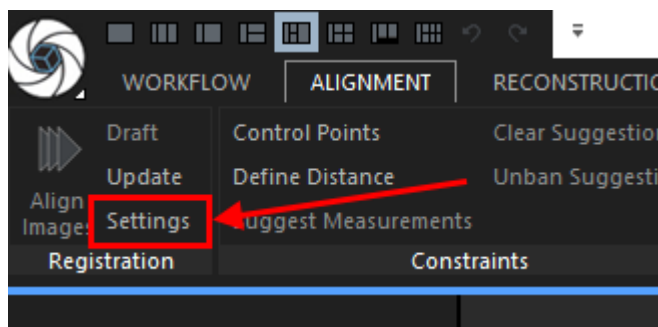
When the alignment process completes, you'll find some number of components (ideally some small number, and ideally ideally where just one contains the majority of the cameras) in your '1Ds' project window:

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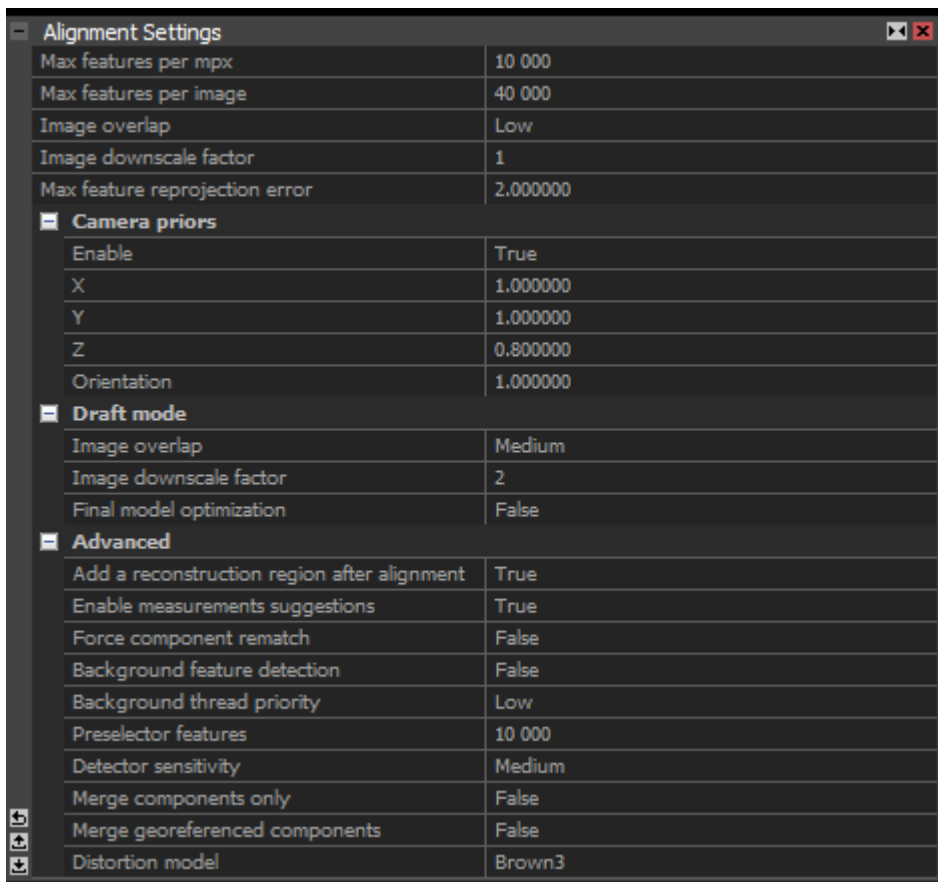


If you're happy with the results from the alignment process, you can continue on to the next step. If not, you'll need to tweak your alignment settings and/or the settings for the inputs that you're attempting to align.

You can change alignment settings by clicking 'Setting' on the 'ALIGNMENT' tab:



Which will pull up this panel at the bottom of the leftmost viewport:

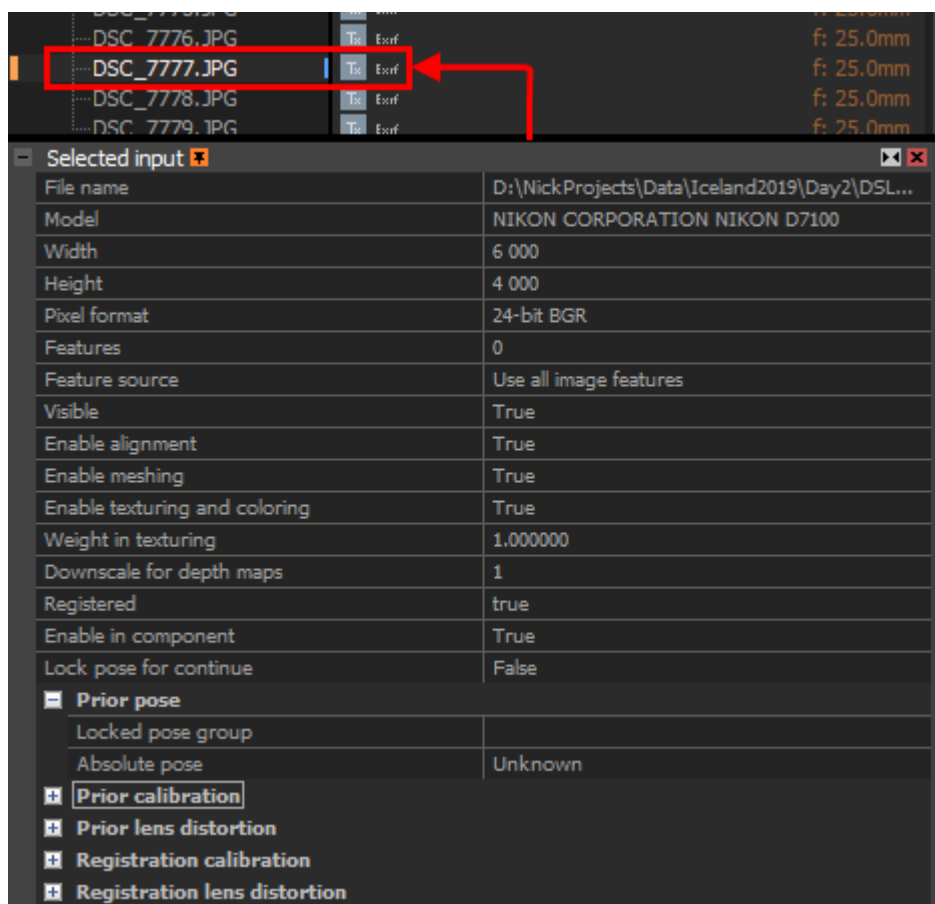


I've found these to be the most useful settings to tweak:

- **Image Overlap:** defines how much 'object overlap' the alignment algorithm should expect an average image to have with its neighbor. A higher number is more restrictive and *should* be faster, but may lead to a bunch of disconnected components. Set to a low number when images share few features - this may lead to some odd or inaccurate alignments.
- **Preselector features:** the number of 'detected features' (max features per image sets a bound on this number) that will be used during alignment. Increasing this number *should* give a better alignment.
- **Force component rematch:** after setting to true, running 'Align Images' again will attempt to find a 'better' alignment based on the cameras that have already been aligned. Think of it as 'optimizing' the existing alignment.

The settings relating to how a given input behaves during alignment can be accessed by clicking on a single input (or shift+click to select multiple) input in the '1Ds' window. When an input is selected, you'll see an 'Input Settings' window that looks like this:

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The most relevant tweakable settings here are:

- Feature source: this determines what features are actually used during the alignment process. There are three options: 'Merge using overlaps', 'Use component features', and 'Use all image features'.
 - Merge using overlaps: RC will align existing components exclusively based on points/images that are shared by those components. This means that, in order to have any affect, two components must share some set of registered inputs.
 - Use component features: When chosen, RC will re-align using only the points used in the alignment of an existing component. This is the default setting for inputs imported as part of an imported component.
 - Use all image features: The slowest process of all these, recommended for a small number of camera poses. The program will use all images/points during alignment.
- Enable Alignment: Set this to 'false' if you'd rather RC's alignment process exclude this input.
- Lock pose for continue: This in conjunction with 'Locked pose group' can be used to maintain the relative alignment of a set of inputs on a re-alignment.
- Locked pose group: This in conjunction with 'Locked pose for continue' can be used to maintain the relative alignment of a set of inputs on a re-alignme

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RC Recommended Alignment Workflow (from Alignment FAQ)

- First try this procedure:
 - i. Delete components with almost no images aligned.
 - ii. If possible, add more pictures.
 - iii. In ALIGNMENT/Settings/Advanced set 'Force component rematch' to True.
 - iv. Run alignment again.
- If the result is not satisfactory enough:
 - i. Delete components with almost no images aligned.
 - ii. Use control points to link the unaligned images together with the aligned ones.
 - iii. Run alignment again.
- If the result is still not satisfactory enough:
 - i. Delete components with almost no images aligned.
 - ii. Delete the control points with too high discrepancy (red numbers).
 - iii. In ALIGNMENT/Settings/Advanced set 'Merge components only' to True.
 - iv. Run alignment again.

Attempted but Failed Alignment Tinkering Workflows

- In separate projects, import laser scan data with each other set of data (e.g. drone images taken on a specific day at a specific altitude, hand cam data taken on a specific day) and align all inputs. Once you have good alignment in all of your projects, export a component from each and import them all into a single project. Now you should have a bunch of components, all with some set of images individually aligned with a single set of laser scan data. Now, set the 'Feature source' of all inputs 'Merge using overlaps' - which should align all components based exclusively on the images that are registered in each (i.e. will align using only the laser scan data).
 - This failed simply because I couldn't get the laser scans to align with all other image sets.
- Reduce the 'Image overlap' to 'Low', and decrease the number of 'Preselector features' to something small.
 - This resulted in few components, but alignment of the images in those components was obviously very wrong (by visual inspection).
- Manually align separate components using control points.
 - I was able to get what appeared to be good alignment of unaligned components using control points, but meshing the merged component was a mess - lots of obvious topographical errors.

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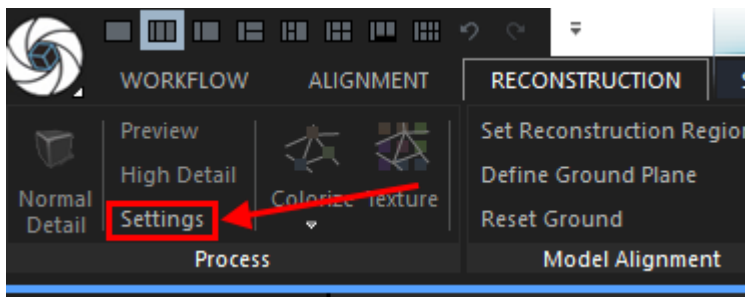
Reconstruction

Before we begin, a few important notes:

- Everything we do during the 'Reconstruction' step relates to whatever component you have selected in the 'IDs' view - so be mindful of which component that is.
- Any reconstruction we do will occur only within the 'Reconstruction Region'. We can have RC automatically create one, or we can create our own. If we create our own, we can alter it using a visual gizmo by selecting the reconstruction region with the mouse and altering it with the handles that appear, OR we can set precise numbers by clicking the 'Info Panel' button in the 'Tools' section of the Reconstruction ribbon.

Now, on to reconstruction!

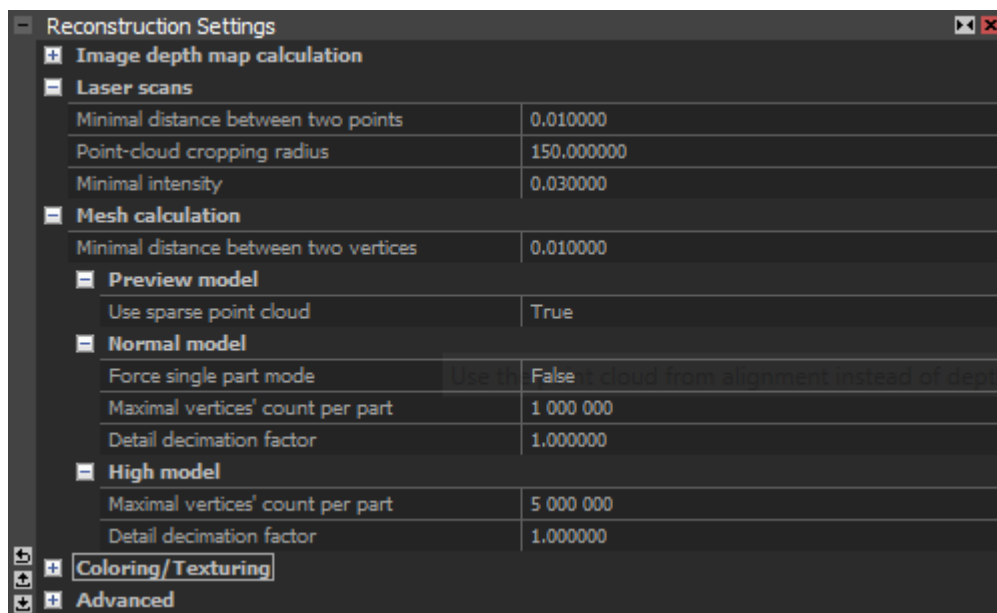
During the 'Reconstruction' step, we'll be creating both a mesh and a texture for the component we've just finished made (in the 'alignment' step). In theory, we can just click two buttons to do this (the 'Normal Detail' button followed by the 'Texture' button on the 'RECONSTRUCTION' Tab), but that likely won't give us the best results. Instead, we'll start by tweaking some of the default reconstructions settings. To open the 'Reconstruction Settings' window, click the 'Settings' button near the bottom left of the 'RECONSTRUCTION' tab/ribbon:



Creating a Mesh

The sections of the 'Reconstruction Settings' panel relevant to mesh creation are 'Laser scans' and 'Mesh calculation':

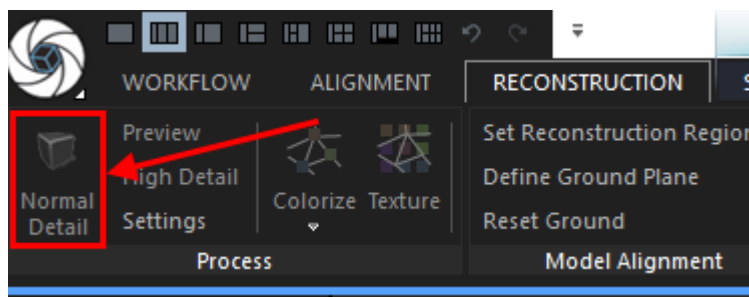
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With the most important fields being:

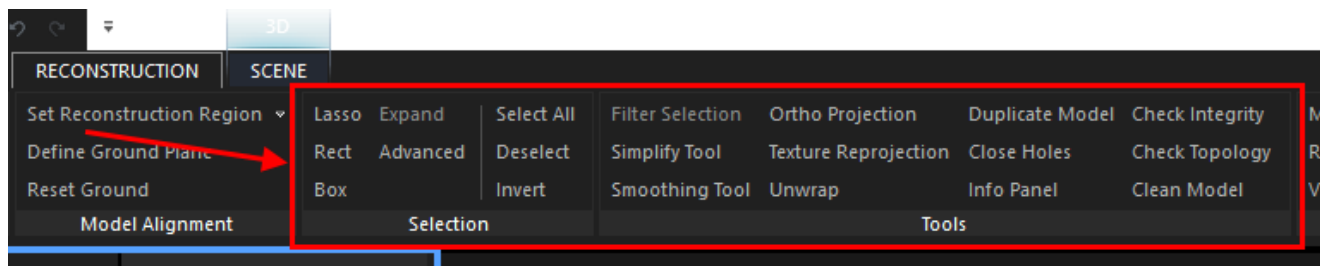
- Minimal distance between two vertices: This sets a minimum bound on the distance any one mesh vertex can be from any other mesh vertex. Setting this to a larger number both makes the meshing process faster and also prevents the creation of excessively small triangles. Set to too high a number, though, and you'll likely lose some detail in the final mesh.
- Force single part mode: Set this to 'True', and RC will create a single mesh for our selected component. If set to 'False', RC will create however many meshes it likes, which can be a LOT. Mesh will generally look quite good, though. As far as I know, there is no way to generate textures for individual mesh parts, though you *can* export all of the parts as separate meshes.
- Maximal vertices count per part/model: Like the name suggests, sets an upper bound on the number of vertices in a part (if multi-part) or model (if single part). If single-part, this can be used to alter the face density of the final mesh. If multi-part, changing this only seems to affect the number of parts created (overall face density remains the same).

Once you have the settings you want, click the 'Normal Detail' (or 'Preview' or 'High Detail', depending on what you want) button to kick off the mesh creation process.



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Once RC finishes creating a mesh, we can edit or clean up that mesh using the options in the 'Selection' and 'Tools' sections of the Reconstruction tab:



Two tools that are particularly useful:

- Select marginal triangles: found in the window that opens after clicking the 'Advanced' in the selection section, this will select all triangles that were added to create an enclosed mesh. We want a single layer mesh, so selecting and removing these triangles is always good.
- Filter Selection: This will create a copy of the current mesh with the currently selected triangles removed - essentially a 'delete' tool with built in versioning.

Creating a Texture

Once we've got a model we're happy with, it's time for texture creation. The 'Coloring/Texturing' portion of the 'Reconstruction Settings' window looks like this:

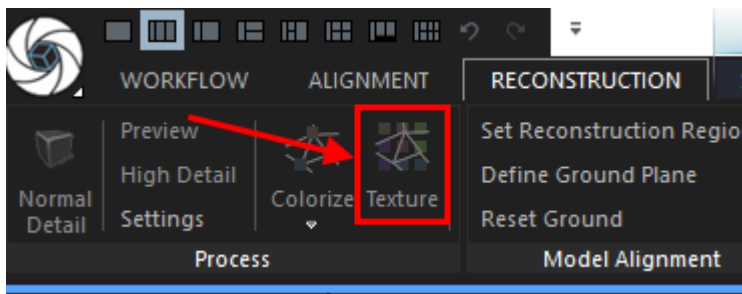


The settings I've found useful to tweak are these:

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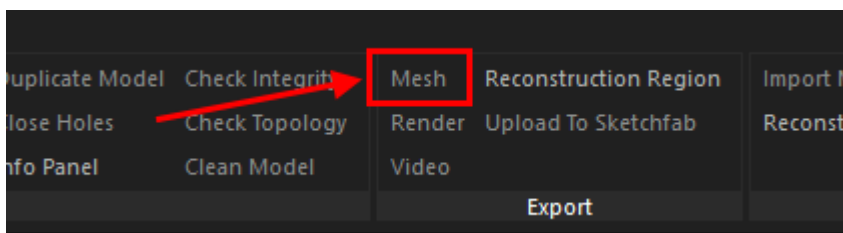
- Maximal texture resolution: the largest a single texture of the mesh can be. RC will create some number (possibly 1, possibly more) of textures for a given mesh, with the number changing based on the texturing settings.
- Large triangle removal threshold: triangles that RC deems to be 'large' will be colored with a single texel - so adjust this number until all the triangles you'd like to have properly textured are below this limit (you can 'select large triangles' using a tool in the 'Advanced selection tool' window).
- Style: Change this to choose between several completely different texture creation regimes:
 - Maximal textures count: just set a max number of textures that can be created. Directly influences the overall fidelity of the texture.
 - Fixed texel size: allows you to set texel size based on some percentage of 'optimal'. Texels all cover same area on your mesh.
 - Adaptive texel size: allows you to set max and min texel size based on some percentage of 'optimal'. Texels may represent different areas on the final mesh.
- Maximal textures count: upper bound on the number of textures that RC can assign to the mesh.

Once we have our settings how we want them, click the 'Texture' button to begin the texture creation process:



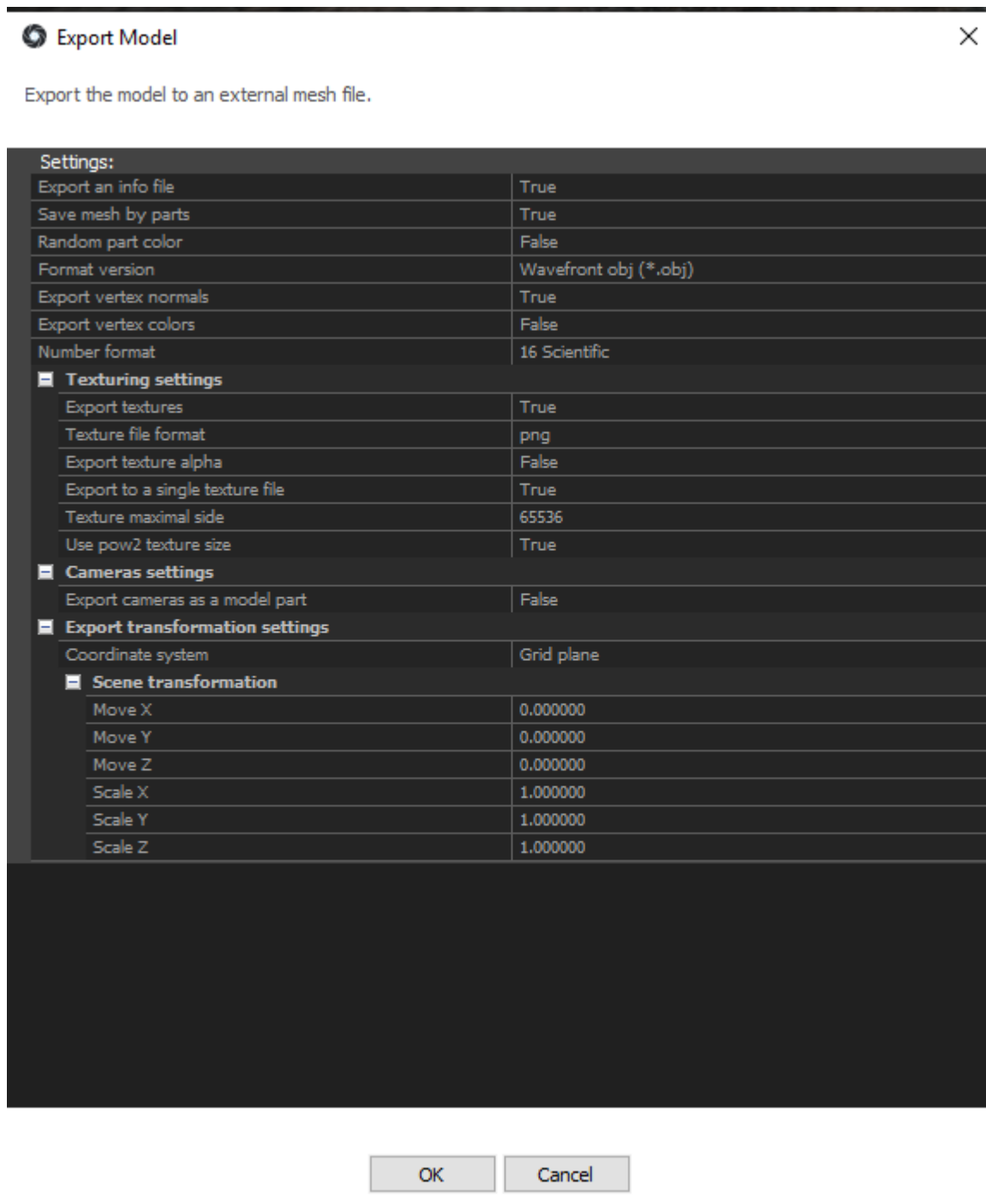
Exporting

With a good-looking textured model, we're now ready to export (finally) our finished product! To do this, select the mesh you want, then click the 'Mesh' button in the 'Export' section of the Reconstruction tab:



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Select whatever options you'd like from the 'Export Model' dialogue and click 'OK':



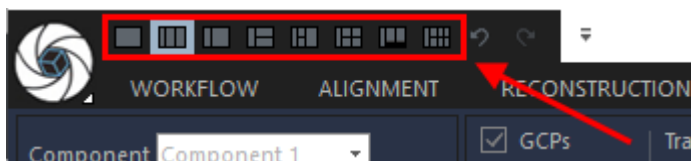
And that's it - we're done! Great Success!

Reality Capture Tips

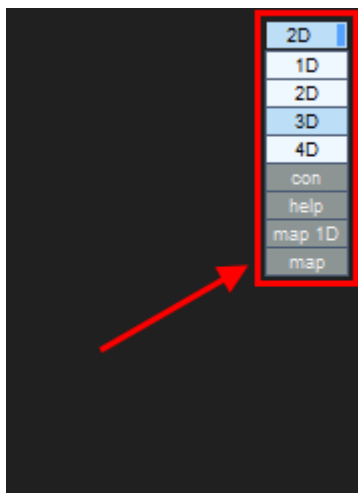
Window Layout

Unlike a bunch of other similar 3d software, RC does *not* allow you to create a purely customized window layout. What it does do is allow you to pick from a bunch of different presets. You can switch between presets by clicking any of the boxy-look icons in the upper left corner of the app:

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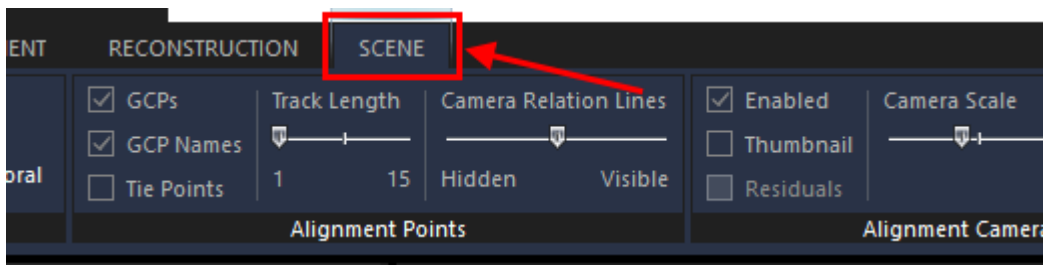
The displayed contents of any window in any of the layouts can be switched between a number of different options. To switch between content modes, click the little button in the upper left corner of a window and select an option from the dropdown:



A slight oddity with displayed contents: every window in any given layout (other than the 1 window layout) is the same *except* for the leftmost window. This window is something akin to the 'project' or 'hierarchy' window in Unity, and generally displays info about the files used by your project or the components that exist in your scene.

Contextual Tab/Ribbon

RC has three static tabs ('WORKFLOW' 'ALIGNMENT' and 'RECONSTRUCTION') and one tab that changes depending on the window you have highlighted - which I'll be calling the 'Contextual Tab' (it should be noted that a second dynamic tab - the 'PROGRESS' tab - also exists; it shows up when you minimize any in-progress process). When a window that has contextual options is selected, the contextual tab will appear to the right of all of the other tabs:



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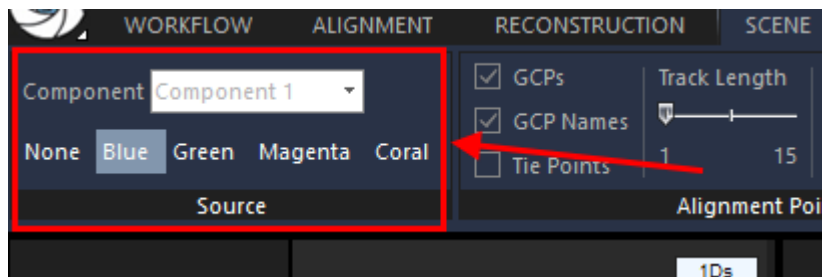
If you want to change a setting that possibly has something to do with a window, recall that this feature exists and be mindful of which window you have selected.

What the Heck Are All These Colors?

You may notice that the 'content display' button in the upper right corner of some windows has a little color next to it. The heck is this thing?



Turns out this has to do with a 'Source' feature of RC. Whenever you select any input from any location, you will be default change the selection of the 'blue' source. If you hold the 2, 3, or 4 keys while selecting an input, you'll change the 'green', 'magenta', and 'coral' sources, respectively. Essentially, these different color sources are a multi-selection feature of RC. To change the color of the source you're viewing in a given window, go to 'Source' section of the contextual tab and select whatever color you'd like:



Super pro tip: while you have (1) a 3D window selected, (2) a component selected, and (3) an input registered to that component selected, hitting the 'enter' key will move the 3D camera view to look through that of the camera corresponding to the selected input.

Components

What the heck even is a component? According to the docs, a component is "a part of a model - a group of images aligned together." It's the core element around which you'll actually be constructing meshes and textures. Once you have a component you're happy with, most of the hard work is done. When exporting or importing components, you'll be manipulating .rcalign files. These files contain all important aspects of a component - such as image references and alignment poses.

Help Window

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One of the window content display modes is the 'help' option. This is basically just built in documentation. It has a search feature and also a number of 'getting started' and 'tutorial' documents. If you're stuck, this is a great first place to look!

RealityCapture Help

You can use the resources in this topic to learn how to automatically convert photos to a 3D model, filter laser scans, and create 3D meshes from laser point clouds, re-color/texture laser scans using images and how to reconstruct scene parts which were invisible to laser scanners and more.

Search

Define Your Goal:

Quick Start	Get Familiar with the Application Basics	Create a 3D Model from Laser Scans	Combine Photos & Laser Scans
Create a 3D Model from Aerial Data	Video as an Input	Create and export an Ortho-Photo & DSM	Fast Draft Reconstruction
Coloring and Texturing	Texture Reprojection	Simplify	Filter
Alpha	Mesh	Render	Merge

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